



Mikko De Torres

Mechatronics, robotics and AI engineer

Mechanical design CAD 3D printing Robotics Machine learning

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Engineer, and the person who explains it

I design mechanical systems and then build them. My work runs from CAD modelling and 3D printing through electronics, control code and testing, and into the documentation and video that explain it.

Seven years in industry as a maintenance and automation engineer, then seven teaching robotics, control systems and machine learning at Batangas State University. I am finishing an MS in Artificial Intelligence on motion planning for six-axis manipulators.

What I bring to a team

A working knowledge of the whole loop: design it, build it, wire it, program it, test it, then document it well enough that someone else can pick it up. Teaching made the last part second nature.

Mechanical design and CAD

SolidWorks, Onshape, AutoCAD. Designed the structure and folding mechanism of a DOST-funded solar power station, built and deployed as a working prototype.

Electronics and code

Arduino, ROS, C++ , Python, MATLAB. Built a 3-DOF manipulator controlled through ROS and Arduino, with the kinematics solved in Python.

Teaching and documentation

Robotics 1 and 2, control systems, machine learning and data science, microprocessors, physical systems modelling, basic workshop and machining. I run a YouTube channel on the same subjects.

Education

BS Mechatronics Engineering, 2012. MS Artificial Intelligence, in progress. Both at Batangas State University.

Certifications

TESDA NC II, Mechatronics Servicing.
BOSH Safety Officer 2 (SO2), 2026.



Collapsible solar power station for farms

A portable, foldable solar power unit for farm use, funded by DOST-TAPI and built to a working prototype.

What I built

The mechanical structure and the folding mechanism, modelled in SolidWorks. The unit carries photovoltaic panels and a battery storage unit, and collapses so one team can move, deploy and store it without special equipment. Linear actuators open and close the panel array.



SolidWorks - open



SolidWorks - front

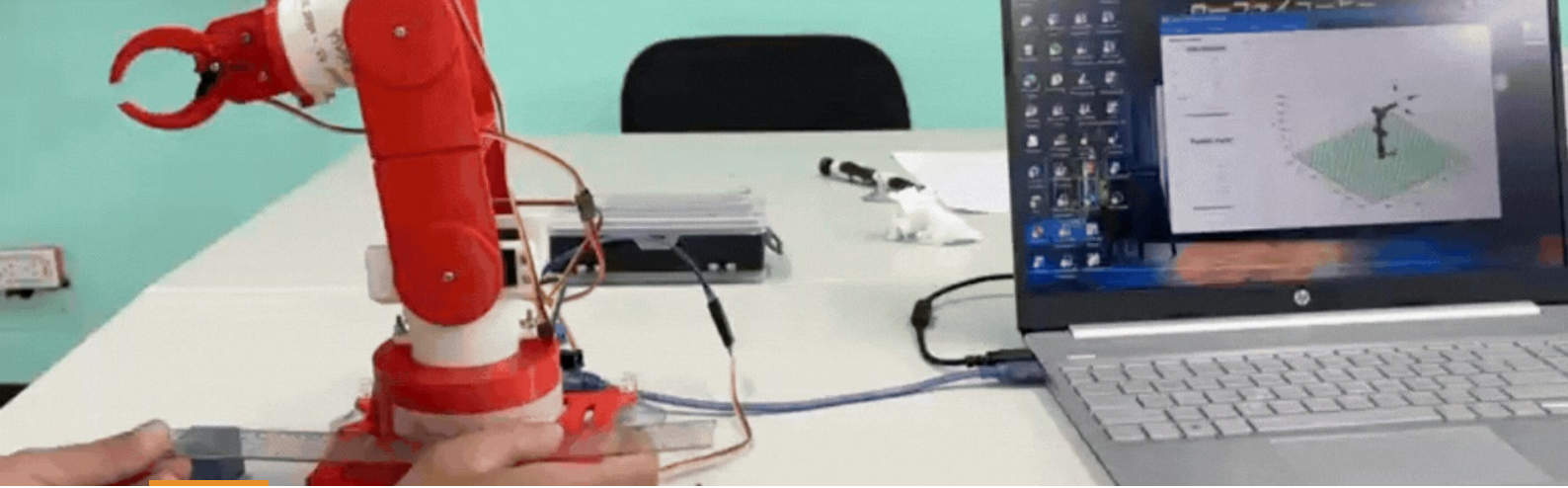


SolidWorks - back

How

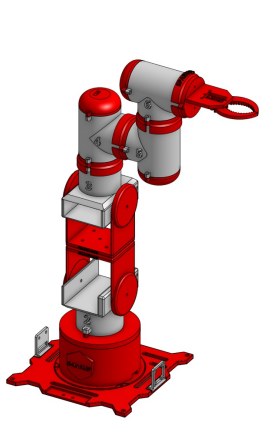
SolidWorks for the structure and mechanism, then fabrication drawings. I also produced the brochure and the instruction materials the users received.

Role Mechanical design lead	Tools SolidWorks, fabrication drawings	Context DOST-TAPI funded project	Year 2023-24
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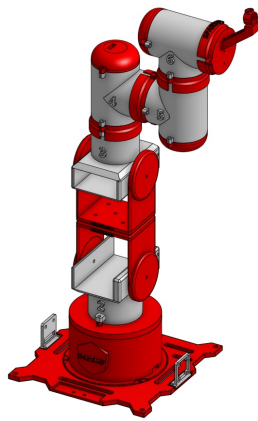


3-DOF to 6-DOF robot arm kit

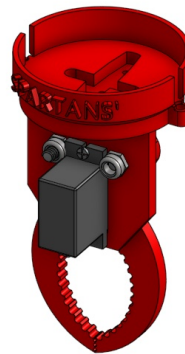
A modular robot arm, 3D printed and assembled from three to six axes depending on the configuration - a teaching platform my students learn kinematics on.



CAD - view 1



CAD - view 2



Gripper design



Assembled arm

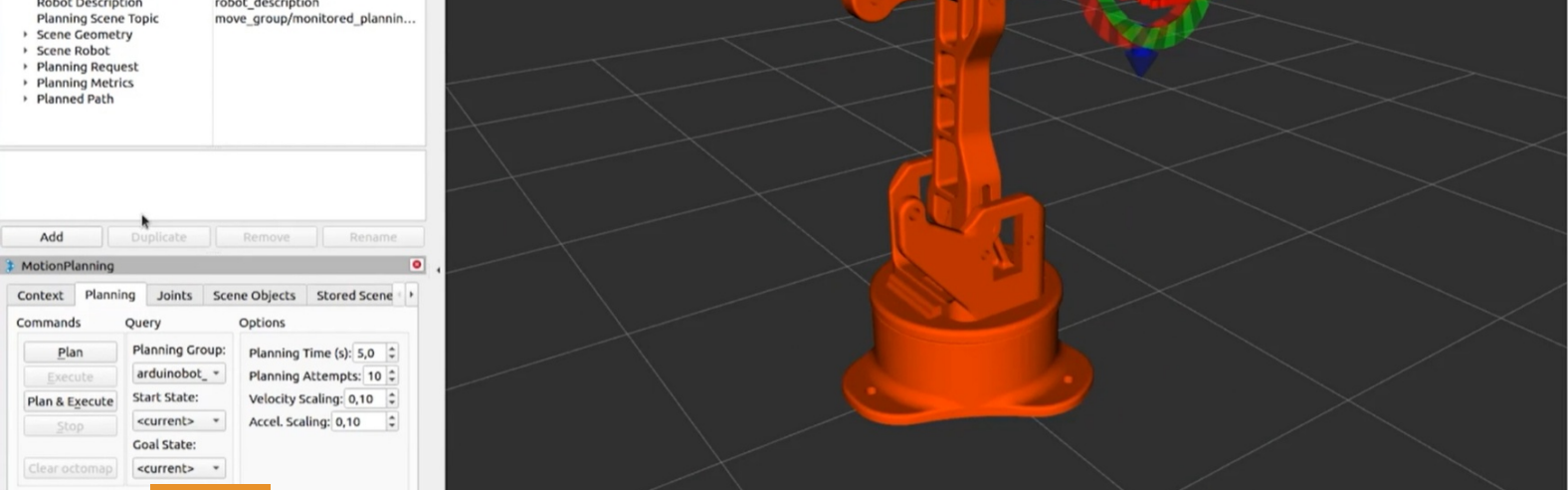
What I built

A six-axis arm designed in SolidWorks and 3D printed. Assembled with servo motors, a custom gripper, and an Arduino-based controller. Used in robotics classes for kinematics, pick-and-place and path planning demonstrations.

How

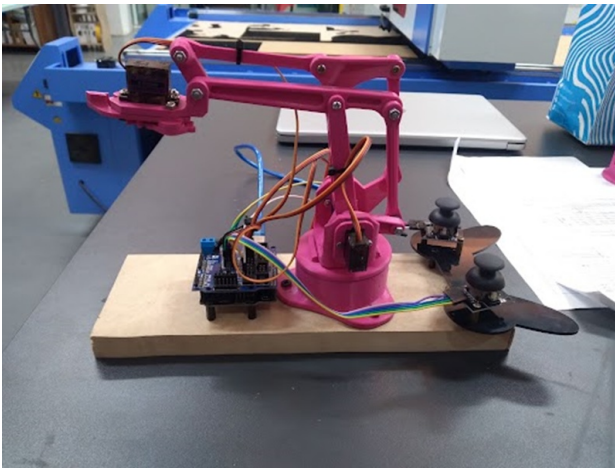
3D printed parts, SolidWorks, Arduino, servo motors. Competed and demonstrated at robotics events.

Role Built and integrated	Tools SolidWorks, 3D printing, Arduino	Context Robotics teaching platform	Year 2023-24
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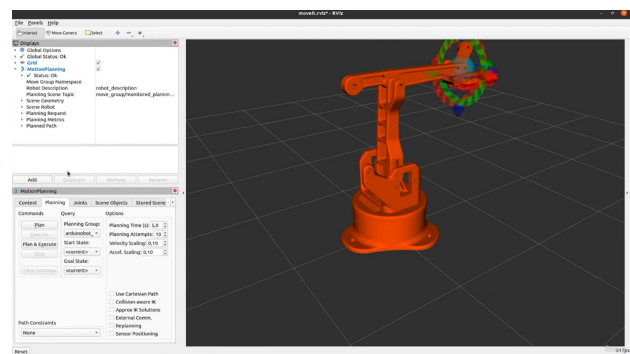


3-DOF articulated manipulator on ROS and Arduino

A three-axis arm where the mechanical build, the electronics and the control software all had to meet.



Actual Arduino robot



MoveIt simulation in RViz

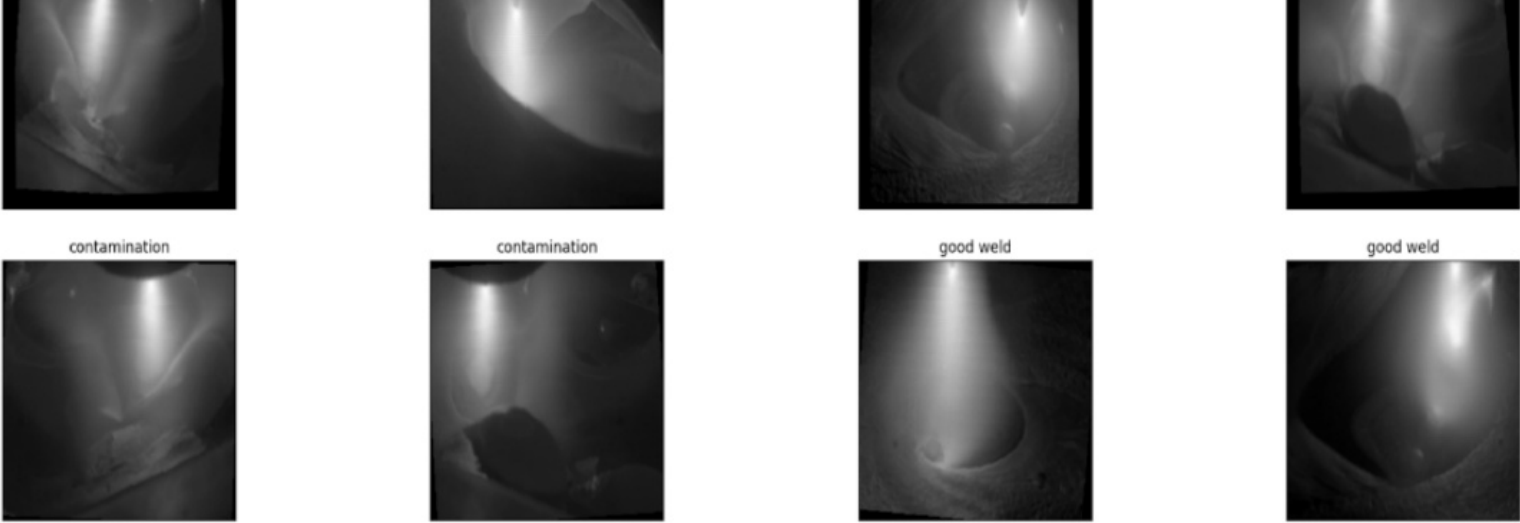
What I built

The manipulator itself, with real-time control written in C++ under ROS and the kinematics solved in Python for motion planning. Simulated in MoveIt for path planning and visualized in RViz.

How

ROS for the control layer, Arduino for the hardware interface, C++ for the real-time loop, Python for the kinematics, MoveIt for motion planning.

Role Built and programmed	Tools ROS, Arduino, C++, Python, MoveIt	Context Robotics laboratory	Year 2023-24
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Weld inspection by machine vision

A vision system that classifies weld quality on stainless steel from HDR camera images - contamination, burn-through, or good weld.

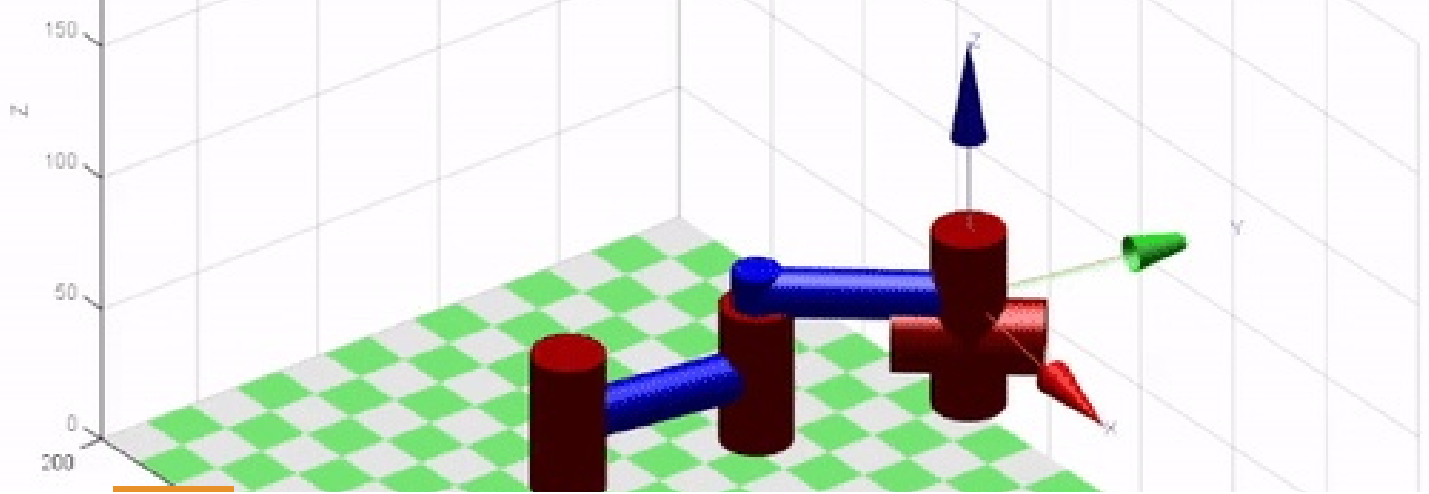
What I built

A full classification pipeline in Python that takes HDR camera images of a weld and classifies the defects. The dataset includes contamination, burn-through and good weld classes, captured under HDR imaging to hold detail across the bright arc and dark surrounding metal.

How

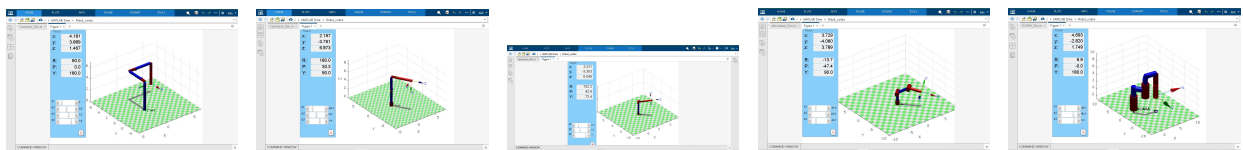
Transfer learning on a pre-trained VGG16 network for feature extraction, then retrained for weld defect classification. HDR imaging holds detail across the bright and dark parts of the weld.

Role Full pipeline	Tools Python, HDR imaging, VGG16, transfer learning	Context Machine vision project	Year 2023-24
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Manipulator simulations in MATLAB

Simulations of every major manipulator type, written for the robotics courses I teach.



Cartesian

Cylindrical

Spherical

Articulated

SCARA

What I built

Simulations of the Cartesian, cylindrical, spherical, articulated and SCARA manipulator types, plus a 6-DOF SCARA V3 (a 3-DOF SCARA with a spherical wrist), each covering kinematics, dynamics and control. Used in Robotics 1 and 2 classes for third-year students.

How

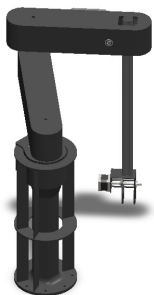
MATLAB with Peter Corke's Robotics Toolbox. Each simulation includes a Teach pendant interface for interactive joint control and end-effector position readout.

Role Wrote all simulations	Tools MATLAB, Robotics Toolbox	Context Robotics 1 and 2 teaching material	Year 2022-24
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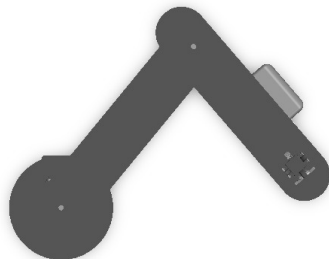


3-DOF RRP SCARA manipulator

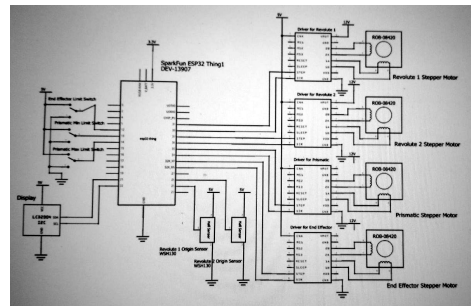
A SCARA-type arm with two rotary joints and a vertical slide. I designed the machine elements and programmed the robot.



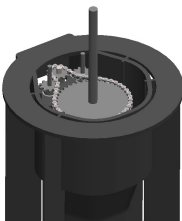
3D model



Top view - reach



Control circuit



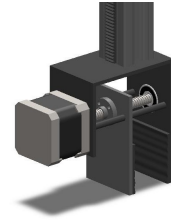
Chain and sprocket



Belt and pulley



Rack and pinion



End effector

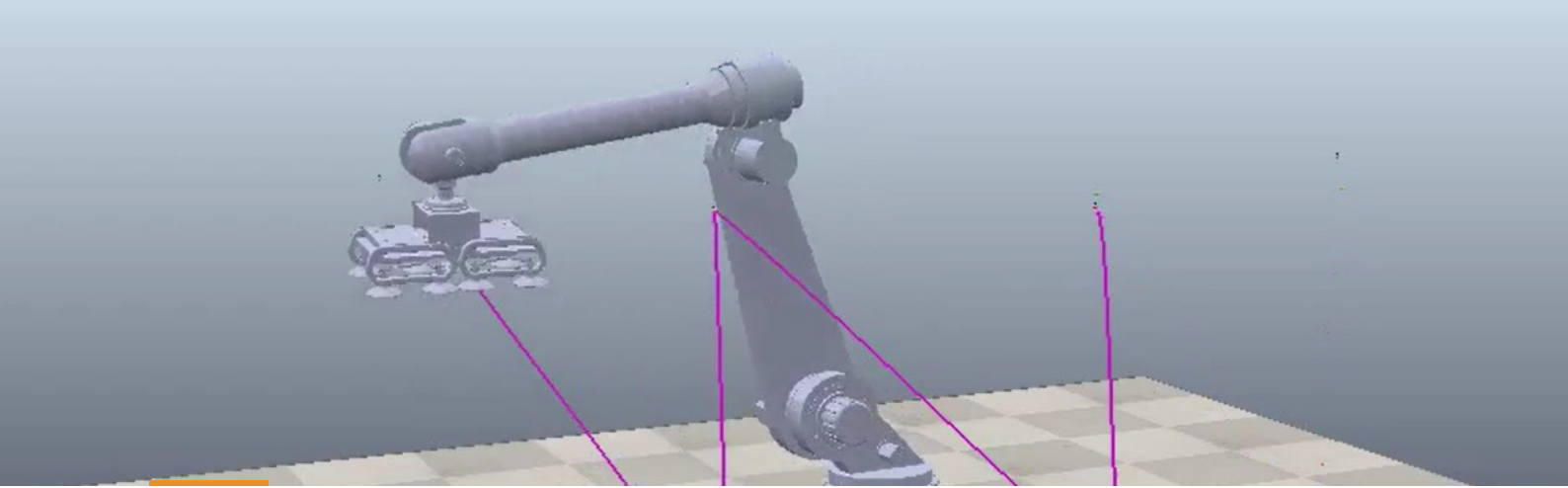
What I built

The drive for each joint: a chain and sprocket in the base, a belt and pulley in the first arm, and a rack and pinion with gear reduction for the vertical slide. The gripper runs on its own stepper motor.

How

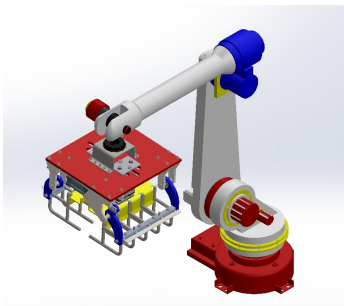
An ESP32 drives four stepper motors. Hall sensors find the home position of the two rotary joints, limit switches stop the slide and the gripper at their ends, and a small LCD shows the readings.

Role	Tools	Context	Year
Design and programming	CAD, ESP32, stepper motors	Robotics Laboratory	2024

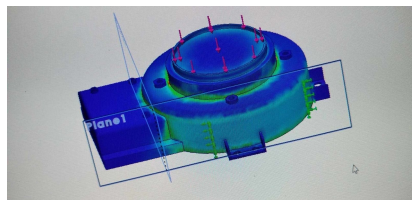


6-DOF manipulator in CoppeliaSim

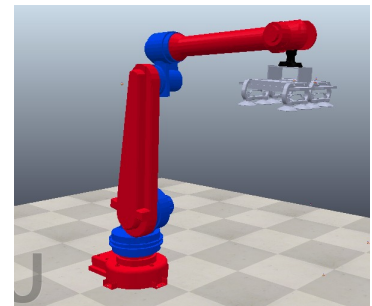
A six-axis arm I designed in SolidWorks, checked for stress, and programmed to move in a CoppeliaSim simulation.



SolidWorks model



Stress study of the base



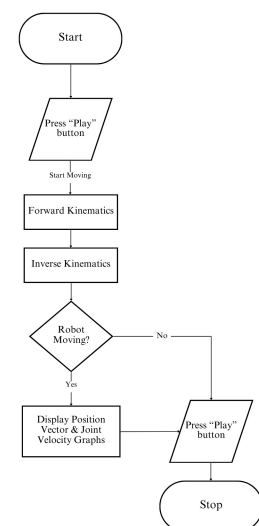
Model in CoppeliaSim

What I built

A six-axis arm modelled in SolidWorks, with a static stress study on the base under load. I brought the model into CoppeliaSim and programmed its forward and inverse kinematics, so the arm follows a set path.

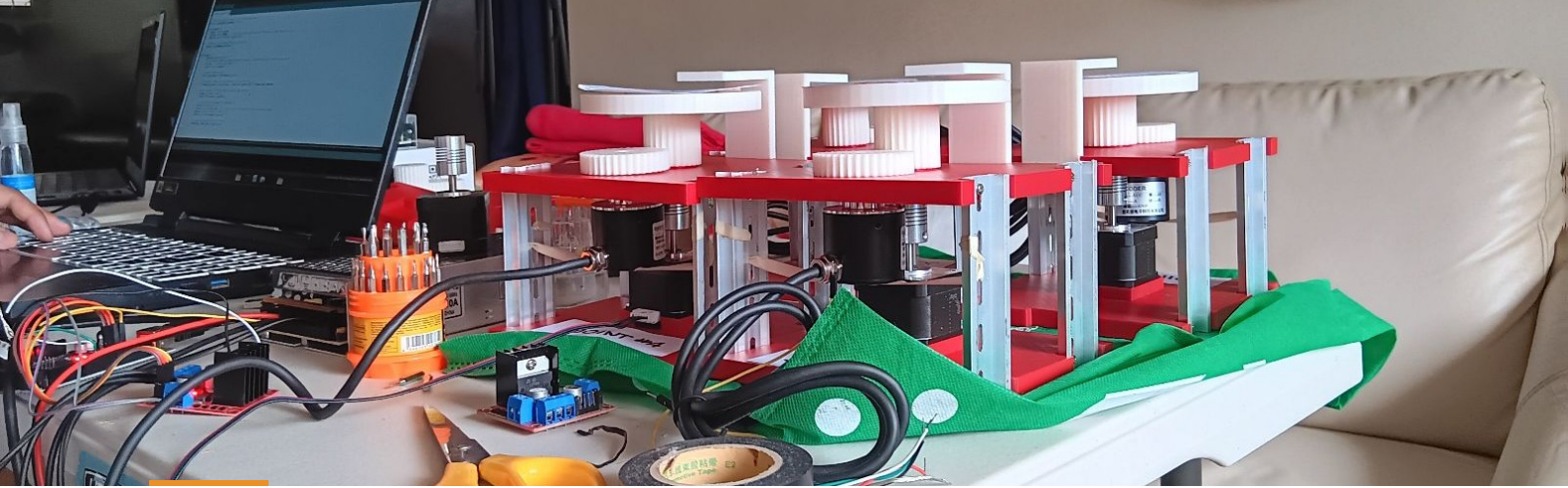
How

Press play and the arm starts moving. While it moves, the simulation plots the tool position and the joint speeds live. SolidWorks for the design and stress study, CoppeliaSim for the simulation.



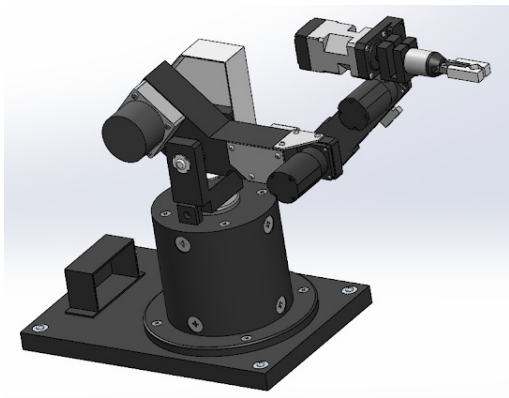
Program flow

Role Design and simulation	Tools SolidWorks, CoppeliaSim	Context Robotics Laboratory	Year 2024
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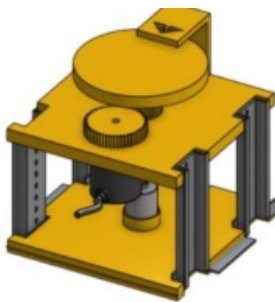


5-DOF articulated manipulator

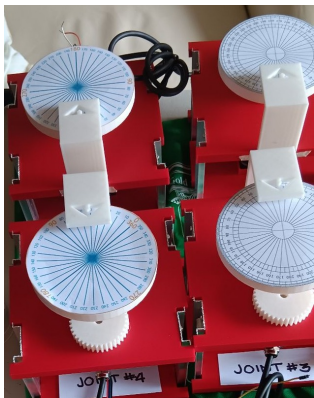
A five-axis arm run through LinuxCNC. I did the real-time testing and the LinuxCNC setup, and helped program it.



SolidWorks model



Joint test rig - CAD



Angle dials on the rigs

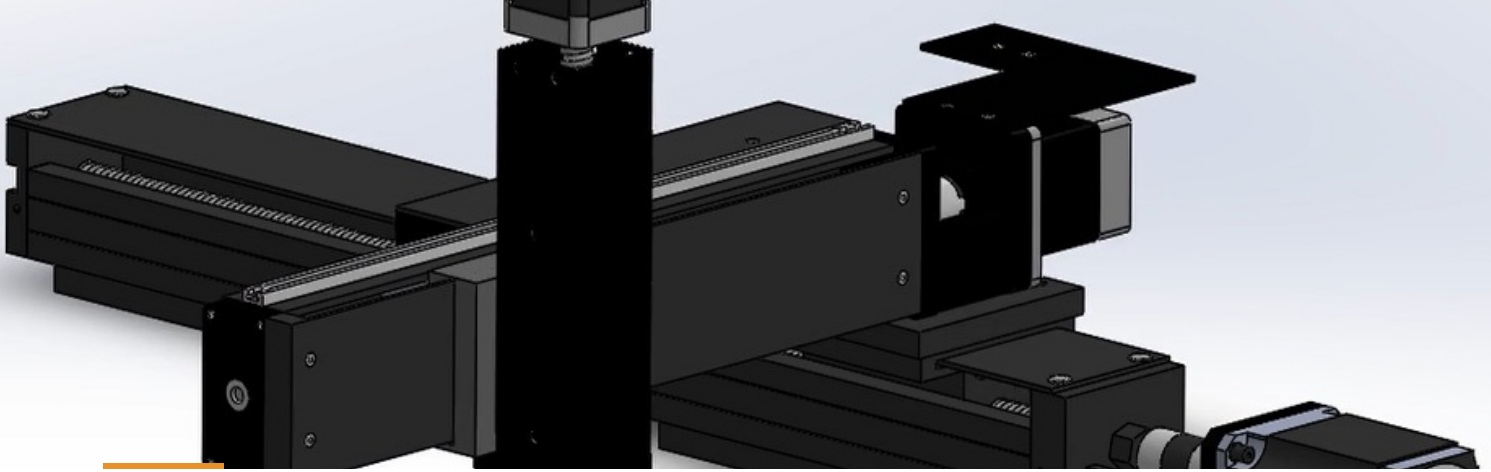
What I built

Each joint was tested on its own rig, with a motor, an encoder, the gear set and a printed angle dial. This let us check each joint’s angle before it went on the arm.

How

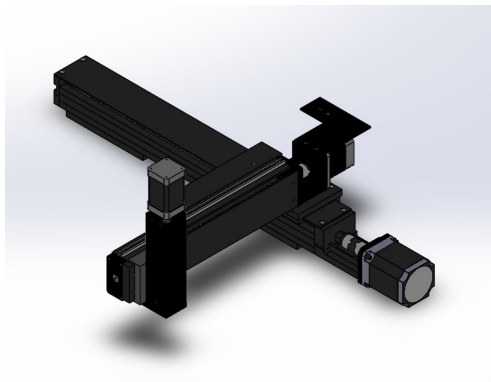
LinuxCNC sends the joint commands to the motors and to a digital twin of the arm, so the model and the real joints move together.

Role Testing and integration	Tools LinuxCNC, motors and encoders	Context Robotics Laboratory	Year 2024
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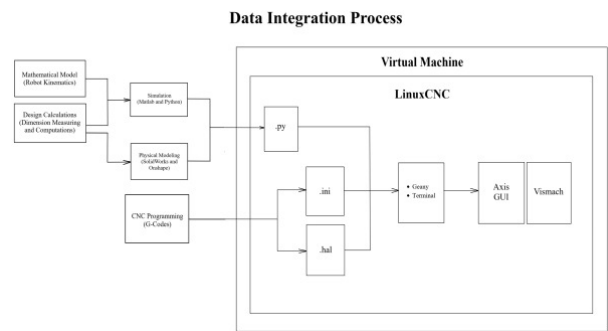


Cartesian CNC robot - digital twin

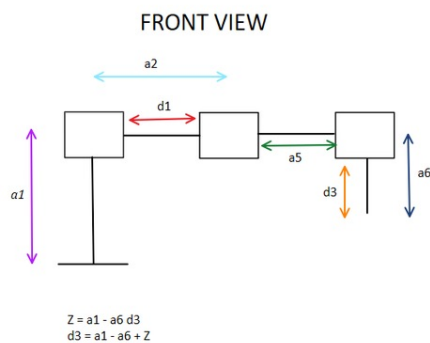
A three-axis Cartesian robot and its digital twin, built with a student team at the Mechatronics and Robotics Laboratory.



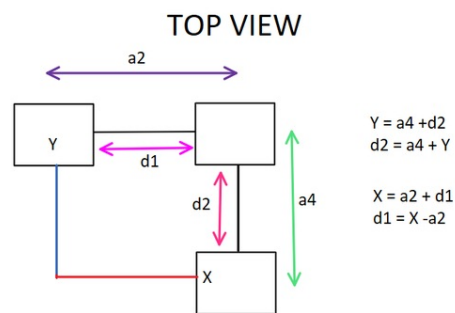
3D model



Data integration



Kinematics - front



Kinematics - top

What I built

The geometric design in SolidWorks and the digital twin, programmed and linked to LinuxCNC. The work covers the kinematics, the data link between the machine and the model, and the engineering drawings.

Role Design and programming	Tools SolidWorks, LinuxCNC, MATLAB	Context Robotics Laboratory	Year 2023
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Instructor | Mechatronics and Robotics Engineer

Teaching, filming and open material

The engineering is only half of it. I also make the material that explains it.



Mikko De Torres

@mikkodetorres · 1.3K subscribers · 136 videos

YouTube - youtube.com/@mikkodetorres

Over 130 videos and 1,300 subscribers. I script, film and edit videos on robotics, data science in Excel, Git and GitHub, and basic workshop and machining.

Course repositories

Robotics_MEXE_3rdYearCourse - kinematics, control algorithms and simulation tasks for Robotics 1 and 2.

MexEE402_AI - data science, machine learning and putting models into use.

Print documentation

Brochure and instruction materials for the solar power station, written for the farmers who would use it.

Role	Tools	Context	Year
Author and presenter	Video, GitHub, technical writing	Public teaching material	2019-present

Let's build something.

If any of this is close to what you are working on, I would be glad to talk it through.

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